

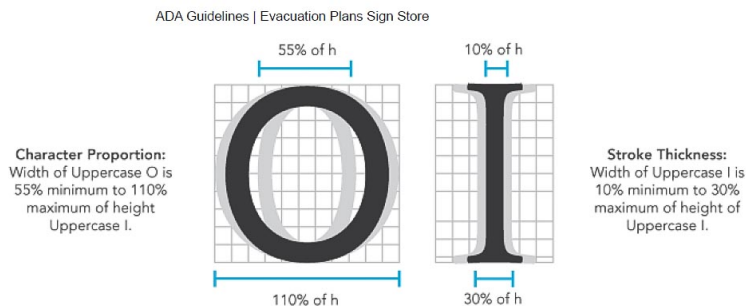
# Tactile Map Making Quick Reference Guide

*All dimensions provided include metric millimeters and inches.*

## Minimum Letter Dimensions for Raised Character Signs

(For tactile characters only; visual requirements separate have parameters)

| Parameter                                         | Dimensions                                                              | Source                           |
|---------------------------------------------------|-------------------------------------------------------------------------|----------------------------------|
| <b>Depth</b> (letter height from background)      | $\geq 0.8 \text{ mm}$ ( $\frac{1}{32}$ " )                              | ADA Signage Guidelines (703.2.1) |
| <b>Character height</b> (uppercase "I")           | $\geq 16 \text{ mm}$ ( $\frac{5}{8}$ " )<br>$\leq 51 \text{ mm}$ (2")   | ADA Signage Guidelines (703.2.5) |
| <b>Stroke thickness</b> (width of uppercase "I")  | $\leq 15\%$ of character height                                         | ADA Signage Guidelines (703.2.6) |
| <b>Character width proportion</b> (uppercase "O") | $\geq 55\%$ of uppercase "O" height to the uppercase letter "I" height. | ADA Signage Guidelines (703.2.4) |

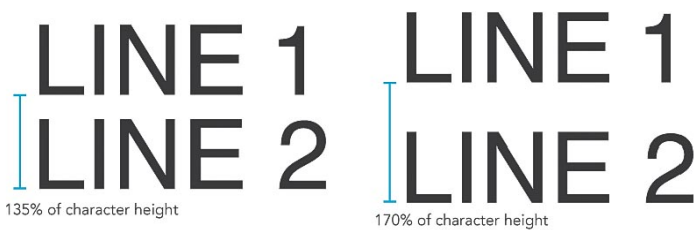


Measurement based on the Uppercase "I"

## Font Style and Spacing

| Parameter                           | Specification                                                                                                                       | Source                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Typeface                            | Sans-serif, uppercase only, no italic/script                                                                                        | ADA Signage Guidelines (703.2.2)         |
| Letter-to-letter spacing            | Closest points between letters: <b>10%–35%</b> of character height <b>and <math>\geq 3.2</math> mm (<math>\frac{1}{8}</math>"</b> ) | ADA Signage Guidelines (703.2.7)         |
| Line spacing (baseline-to-baseline) | <b>135%–170%</b> of character height                                                                                                | ADA/Braille Signage Guidelines (703.2.8) |

SPACING



---

## Spacing Guidelines

| Parameter                                                                                        | Specification                                                | Source                                   |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------|
| Space around tactile letters and symbols (includes borders, braille, raised letters and symbols) | <b><math>\geq 9.5</math> mm (<math>\frac{3}{8}</math>"</b> ) | ADA/Braille Signage Guidelines (703.2.7) |



---

## Finish and Contrast Recommendations

| Parameter             | Specification                                                               | Source                 |
|-----------------------|-----------------------------------------------------------------------------|------------------------|
| <b>Surface finish</b> | Matte/non-glare                                                             | ADA Signage Guidelines |
| <b>Color contrast</b> | High-contrast ( $\geq 70\%$ LRV difference), dark-on-light or light-on-dark | ADA Signage Guidelines |

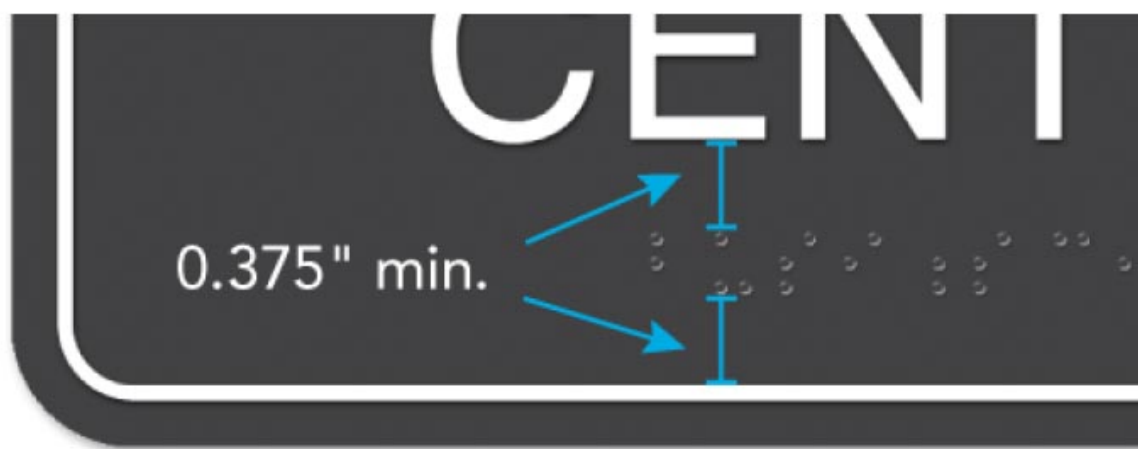
---

## Quick implementation tips

1. **Spacing Consistency:** Keep every doorway gap exactly the same width. Keep lettering and symbols on map and key the exact size.
  2. **Print preview check:** In your slicer's layer view, check wall count matches intended sizing.
  3. **User testing:** Before the final process is saved, get feedback from a tactile map user to be sure nothing was missed and all information can be easily understood.
-

## Braille Reference Guide:

703.3.1 Dimensions and Capitalization. Braille dots shall have a domed or rounded shape and shall comply with Table 703.3.1. The indication of an uppercase letter or letters shall only be used before the first word of sentences, proper nouns and names, individual letters of the alphabet, initials, and acronyms.



## Brennen's 3D Printed Map Quick Reference Guide: Depth

(General change use 0.6 mm difference and 0.8 mm difference for symbols)

| Feature            | Depth  |
|--------------------|--------|
| Braille Base Layer | 0.1 mm |
| General Base Layer | 0.4 mm |

|                             |        |
|-----------------------------|--------|
| <b>Interior Floor Layer</b> | 1.0 mm |
|-----------------------------|--------|

|                      |        |
|----------------------|--------|
| <b>Symbols Layer</b> | 1.8 mm |
|----------------------|--------|

|                          |        |
|--------------------------|--------|
| <b>Outer Walls Layer</b> | 2.4 mm |
|--------------------------|--------|

Test of alternate version on (6\_24\_2025):

- Braille Base 0.1 or 0.7 (0.3mm separation)
- Starting base plate street below 0.4
- Starting base plate garage floor 1.0 (0.6mm depth)
- Sidewalk 1.6 (0.6mm depth)
- Symbols layer 1.2 or 1.8 or 2.4 (0.8mm depth above base)
- Bollards height 3.0 (0.6mm depth)
- Wall Height 3.6 (0.6mm depth)

## **Brennen's 3D Printed Map Quick Reference Guide: Sound Transit Symbol Sizing**

| <b>Feature</b>                                  | <b>Width</b> | <b>Height</b> |
|-------------------------------------------------|--------------|---------------|
| <b>Wall</b>                                     | 0.2 mm       | N/A           |
| <b>Curb</b>                                     | 0.1 mm       | N/A           |
| <b>Shoreline Side</b>                           | 1.5 mm       | N/A           |
| <b>Paid Fare Zone</b><br>(Repeating Line Width) | 0.8 mm       | 16 mm         |

|                                                  |                            |       |
|--------------------------------------------------|----------------------------|-------|
| <b>Platform Edge</b>                             | 1.5 mm                     | N/A   |
| <b>Track</b> (Reference to Curb)                 | 2 x 1 mm (with 1.5 mm gap) | N/A   |
| <b>Turnstiles</b>                                | 6 mm                       | 2 mm  |
| <b>Stairs</b>                                    | 1.2 mm                     | 16 mm |
| <b>Escalator</b>                                 | 1.2 mm                     | 18 mm |
| <b>Elevator Up / Down</b><br>(Outer Line Width)  | 2.2 mm                     | 16 mm |
| <b>Ticket Fare Machine</b><br>(Outer Line Width) | 1.0 mm                     | 16 mm |
| <b>Directional Arrow</b> (Line<br>on arrow)      | 1.2 mm                     | 16 mm |
| <b>Skybridge Number</b>                          | 2.0 mm                     | 16 mm |